

PRABH SINGH

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Professional Summary

Data Engineer with 6+ years of experience building robust, large-scale data pipelines and cloud data platforms (**AWS, OCI**) across healthcare and analytics domains, working with **PySpark, Apache Iceberg, Hive, and Hadoop**.

Expanding into AI engineering with hands-on production work in agentic systems, custom **LLM** tool interfaces, **RAG** pipelines, **NL-to-SQL** engines, and natural language interfaces over live data systems using **Ollama, vLLM**, and fine-tuned **Llama / Mistral** models (**LoRA/PEFT**).

Strong foundation in **Python, SQL**, and cloud platforms combined with growing **MLOps** and AI deployment capabilities (**Docker, Kubernetes, MLflow**) – able to bridge data infrastructure and AI systems end-to-end.

Technical Skills

Data Engineering: PySpark, Apache Hive, Impala, Hadoop/HDFS, Apache Sqoop, Apache Iceberg/Hudi, Delta Lake, Apache Airflow, Apache Kafka, Databricks, Data Modeling, Data Quality & Governance

BI & Analytics: Tableau, Power BI, Microsoft Fabric (Data Models, Lakehouses), Tableau Prep, DAX, Tableau REST API, Power BI REST API

AI / GenAI (Active Projects): Agentic Workflows, Custom LLM Tool & API Integration, Natural Language to SQL, Prompt Engineering, RAG Pipelines, LLM Fine-tuning (LoRA/PEFT), LangChain, MCP (Model Context Protocol)

LLM Serving & Tools: Ollama, vLLM, Hugging Face Transformers, OpenAI API, Llama, Mistral

MLOps: MLflow, Docker, Kubernetes, CI/CD Pipelines, Model Registry, Model Serving, Jenkins, Terraform

Cloud – AWS: S3, EC2, Glue, Athena, EMR, SageMaker, Lambda, Kinesis, Redshift, Step Functions, EKS, ECR, CloudWatch, IAM, Secrets Manager

Cloud – OCI: Autonomous Data Warehouse (ADW), Data Flow (Spark), Data Integration, GoldenGate, Data Science, Model Deployment, Container Registry, Object Storage, OCI DevOps, Functions, Vault, IAM, Terraform

Languages & Tools: Python, SQL, Bash, R, Git, Linux, Jupyter, Conda

Professional Experience

Cotiviti

Sep 2021 – Present

Lead Data Engineer (Feb 2026 – Present)

Pune

Senior Data Engineer | Python, PySpark, GenAI, RAG, vLLM, Tableau

Feb 2024 – Feb 2026

- Designed and shipped 3 production **AI agents** that expose enterprise data systems (**Impala, Tableau, Power BI**, legacy Python) as natural language interfaces for business users.
- Built **RAG** pipelines integrating enterprise data stores with locally-served LLMs (**Ollama**), enabling context-aware query resolution over proprietary document corpora.
- Fine-tuned open-source LLMs (**Llama, Mistral**) using **LoRA/PEFT** on domain-specific datasets; deployed and served models via **Ollama** and **vLLM**.
- Engineered **MLOps** workflows for model packaging, versioning, and deployment using **Docker, Kubernetes**, and **MLflow**-based model registries.
- Architected hybrid on-prem/**OCI** cloud data infrastructure supporting both batch data pipelines and real-time AI model inference.
- Improved **Tableau** dashboard query performance by **200–300%** through **Iceberg**-based data layer redesign and **SQL** optimization.
- Enforced row-level security and column masking on sensitive healthcare datasets, ensuring **HIPAA**-compliant data access for analytics and ML pipelines.

Data Engineer | PySpark, Hive, Impala, Sqoop, Hadoop, Apache Iceberg

Sep 2022 – Feb 2024

- Built and maintained large-scale **PySpark** pipelines for ingestion, transformation, and monitoring of **100M+** record healthcare datasets within the **Hadoop** ecosystem.
- Led migration of Oracle relational data to **HDFS**, standardizing schemas into **Parquet/Avro** formats and reducing processing time by **35%**.
- Adopted **Apache Iceberg** table format enabling schema evolution, time-travel queries, and reliable rollback for regulated healthcare data workflows.
- Designed automated data quality validation pipelines with rule-based checks, reducing downstream reporting incidents by **50%**.

Business Analyst | SQL, Tableau, Tableau Prep

Sep 2021 – Sep 2022

- Designed and delivered 15+ **Tableau** dashboards supporting client engagement, sales operations, and executive reporting.
- Built **SQL** and **Tableau Prep** ETL workflows that reduced manual reporting effort by **60%**, freeing analyst capacity for higher-value work.

- Translated complex data into actionable insights adopted in quarterly sales strategy reviews with senior stakeholders.

ZS Associates

Nov 2020 – Jun 2021

Decision Analytics Associate | Python, PCA, LCA, Logistic Regression

Pune

- Applied **PCA**, Factor Analysis, and **Latent Class Analysis (LCA)** on physician survey data to segment the U.S. gynecologist market into 5 actionable personas.
- Built **multinomial logistic regression** models to project segment membership nationally, directly informing targeted drug promotion strategy for a top pharma client.

HUDL

Jun 2019 – Jul 2020

Sports Analyst | Data Analysis, Reporting

Pune

- Produced data-driven player performance reports used by coaching staff for game-day strategy and athlete development planning.

AI Engineering Projects

Natural Language BI Agent | *Python, Ollama, Tableau REST API, Power BI REST API*

2025 – Present

- Built an **LLM**-powered agent that interprets natural language prompts and autonomously selects the right **Tableau** or **Power BI** dashboard using prompt-engineered API context.
- Agent dynamically constructs API parameters from the prompt, calls the **Tableau** / **Power BI REST APIs**, and returns the correct view or data – eliminating manual dashboard navigation for end users.

NL-to-SQL Analytics Engine with Metric Registry | *Fine-tuned LLM, Ollama, Impala, Matplotlib* 2025 – Present

- Designed and built a multi-stage **NL-to-SQL** system: a fine-tuned **LLM** translates user prompts into a structured **JSON** intermediate representation, which is passed to a custom SQL generator.
- Built a **Metric Registry** – a curated catalog of business metrics with dimension/measure definitions and relationship mappings – that the SQL generator uses to produce semantically correct, complex business queries (multi-table joins, unions, conditional logic) against **Impala**.
- Fine-tuned a second **LLM** to analyze query result structure and automatically select the appropriate chart type; renders final visualizations using **Matplotlib**, delivering a full prompt-to-chart pipeline without manual SQL or BI tooling.
- Engineered a **Python REST API** layer to bridge a distributed infrastructure: LLM inference running on **OCI**, **Impala** cluster on on-premises servers, and a web UI hosted on-prem – designed this way to satisfy data security requirements while keeping model compute in the cloud.

LLM Interface for Legacy Synthetic Data Engine | *Python, Ollama, REST API, JSON Schema*

2025 – Present

- Designed an **API** layer on top of a legacy **Python** synthetic data generation codebase, exposing its configuration as a structured **JSON schema**.
- Prompt-engineered** an **LLM** to interpret natural language data generation requests and produce valid **JSON** configs that are passed directly to the legacy engine – enabling non-technical users to generate synthetic datasets without touching underlying code.
- This pattern mirrors the **Model Context Protocol (MCP)** design: LLM as orchestrator, structured schema as the tool interface, legacy system as the executor.

Education

Great Lakes Institute of Management | The University of Texas, Austin (Online) Dec 2019 – Nov 2020

Post Graduate Program in Business Analytics & Business Intelligence

Bharti Vidyapeeth College of Engineering, Pune

Jul 2015 – Jul 2019

Bachelor of Technology in Mechanical Engineering

Bharti Vidyapeeth College of Engineering, Pune

Jul 2015 – Jul 2018

Diploma in Network Security

Certifications

- Oracle Cloud Infrastructure – Data Platform Associate (1Z0-1195-25) 2025
- Oracle Autonomous Database Professional (1Z0-931-25) 2025
- Post Graduate Program in Business Analytics – Great Learning 2020